

Valentina Eppers

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Education

Bachelor of Science in Environmental Biology

Aug 2023 – May 2027

CALIFORNIA STATE POLYTECHNIC UNIVERSITY | *POMONA*

Research Experience

Scordato Lab, California Polytechnic University Pomona Department of Biology

Feb 2024 – Present

- Helped develop a new analysis protocol using ImageJ to measure variation in a plumage trait involved in signaling behavior in birds distributed throughout Southeast Asia.
- Filtered and aligned whole-genome sequence data using bwa and fastqc.
- Population structure analysis using Admixture and PCA.
- Calculated F_{st} , π , and dxy using pixy to identify patterns of selection in oceanic swallows using whole genome sequence data.
- Experience phasing using Beagle 5.5.
- Used the rehh package in R to calculate XP-EHH and iHS in migratory and non-migratory populations of welcome swallows.
- DNA extraction, data organization, collection, and entry.
- **Collaborative research project:** Identifying genomic regions under parallel selection for commensal nesting behavior in Oceanic swallows.
- **Independent research project:** Genome-wide divergence between migratory and non-migratory welcome swallows (*Hirundo neoxena*).

Organization for Tropical Studies Summer Program in Costa Rica

July 2024

- Introduction to the fundamental principles of tropical biology and the natural history of tropical plants and animals in Costa Rica.
- Conducted independent and collaborative research projects at three field stations, collecting data out in the field, and presenting our results.
- **Independent research project:** Seed preference and predation between native and non-native plant species by Central American agoutis (*Dasyprocta punctata*).
- **Collaborative research projects:**
 - The influences of *Socratea exorrhiza* root “seed cages” on seed predation and seed movement by terrestrial mammals.
 - *Ctenosaura similis*: how size and habituation influences stimuli response.

Presentations

April 2025

- **Identifying genomic regions under parallel selection for commensal nesting behavior in Oceanic swallows.** Poster presentation at California State Polytechnic University, Pomona College of Science Research Symposium

June 2025

- **Genome-wide divergence between migratory and non-migratory welcome swallows (*Hirundo neoxena*).** Poster presentation at the annual meeting of the Society for the Study of Evolution, Athens, GA.

March 2026

- **Genome-wide divergence between migratory and non-migratory welcome swallows (*Hirundo neoxena*).** Oral presentation at California State Polytechnic University, Pomona RSCA Conference.

Awards

- Recipient of Undergraduate Diversity at Evolution (UDE) Fund 2025
- Dean's List 2023-2025
- Presidents List 2025

Relevant Classes

- Foundations of Biology and Laboratory, Biostatistics and Biostatistics Laboratory, Principles of Ecology and Ecology Laboratory, Principles of Evolution, Microbiology and Laboratory, Genetics, GIS, Biology of Fishes, Biodiversity Conservation, Marine Biology, Mycology, Statistics for biologists using R, Tropical Marine Field Ecology.

Volunteer Work

Orange County Bird of Prey Center

Nov 2021 - Feb 2025

- Cleaning enclosures and feeding the wild birds of prey being rehabilitated at the center and learned proper raptor handling techniques.
- Aided in releasing wild birds, assisted with educational outreach, and helped set up events.

Cats In Tow

January 2019 - Present

- Cleaning the cat center, talking with potential adopters, and helping with adoption events.
- Administering medications, feeding, and socializing cats for adoption.